Advances in Real-Time Rendering in 3D Graphics and Games

Creating the Atmospheric World of Red Dead Redemption 2

A Complete and Integrated Solution



thrive
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Rockstar Games



Video



Motivation

- Game takes place at turn of the century
- Undeveloped wilderness
- Several distinct climate regions
- Focus on natural lighting
  - Sky as main contributor
- Sky rendering integral part of RDR1
  - Take it to next level for PS4/XB1





Motivation

Inspired by many American landscape painters of that era



Albert Bierstadt
Passing Storm over the Sierra Nevada, 1870



Asher Brown Durand
Scene from "Thanatopsis", 1850



Frederic Edwin Church
Rainy Season in the Tropics, 1866

Motivation

Volumetric effects are an integral part of the scene



Albert Bierstadt
Storm in the Rocky Mountains, Mount Rosalie, 1866



Frederic Edwin Church
Twilight in the Wilderness, 1860

Motivation



Albert Bierstadt
Among the Sierra Nevada, California, 1868



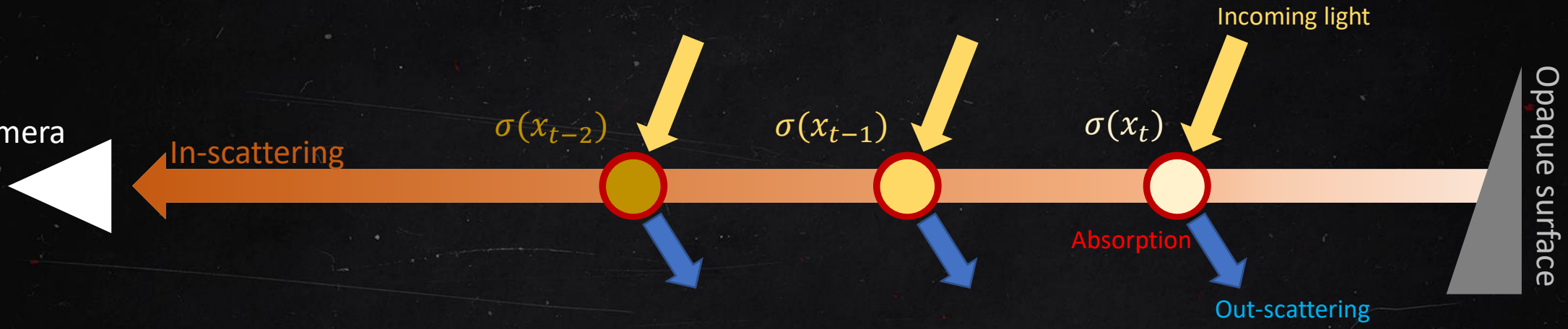
Albert Bierstadt
Scenery in the Grand Tetons, 1869



Asher Brown Durand
The First Harvest in the Wilderness, 1855

Motivation: Physical Basics

Volumetric single scattering



Scattered Light: $SL(x) = \sum_{i}^{|lights|} P(g, \theta_i)\, V(i,x) L(i,x)$

Transmittance: (Beer's Law) $T(a,b) = e^{-\int_{x=a}^{b} \sigma_{ext}(x)dx}$

Extinction: $\sigma_{ext}(x) = \sigma_{abs}(x) + \sigma_{scat}(x)$

Integration:

$$Li(a,b) = Li(b)T(a,b) + \int_{x=a}^{b} SL(x)\sigma_{scat}(x)T(a,x)dx$$

Design Goals



- Unified volumetric rendering system
  - Single lighting and data model
  - Cover various volumetric materials
- Cover large and small scale effects
  - Seamlessly (!) support interiors and exteriors
  - NO POPPING!
- Physically based
- Scalable for low and high end platforms

Previous Work



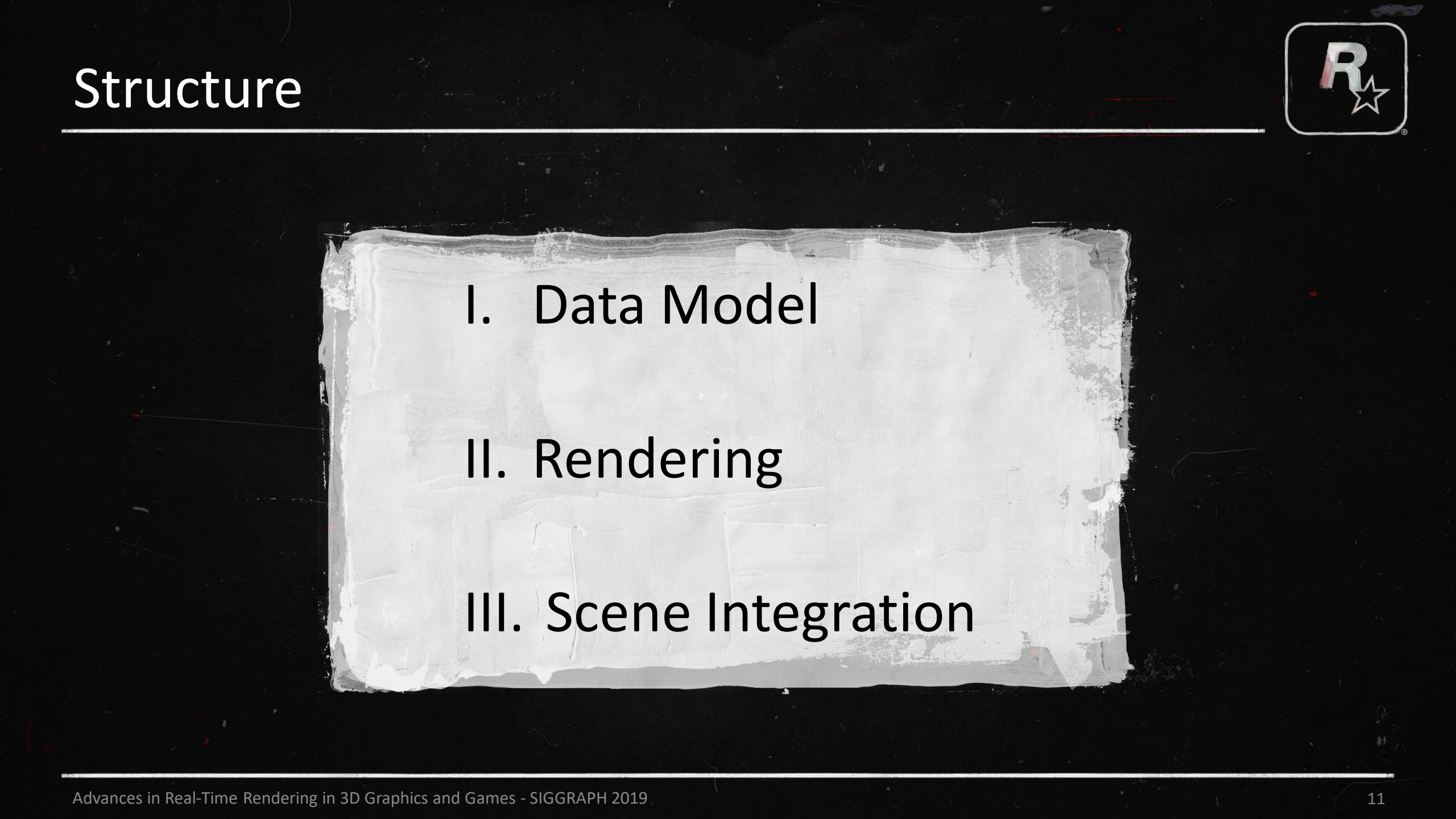
- Analytical Models [Wenzel07]
- Raymarched Fog [Toth09] [Glatzel14]

- Volumetric Fog [Wronski14]
- PBR Froxels [Hillaire15]

- Raymarched cloudscapes [Schneider15]

- Atmospheric scattering models [Bruneton07][Tatarchuk15]

Structure

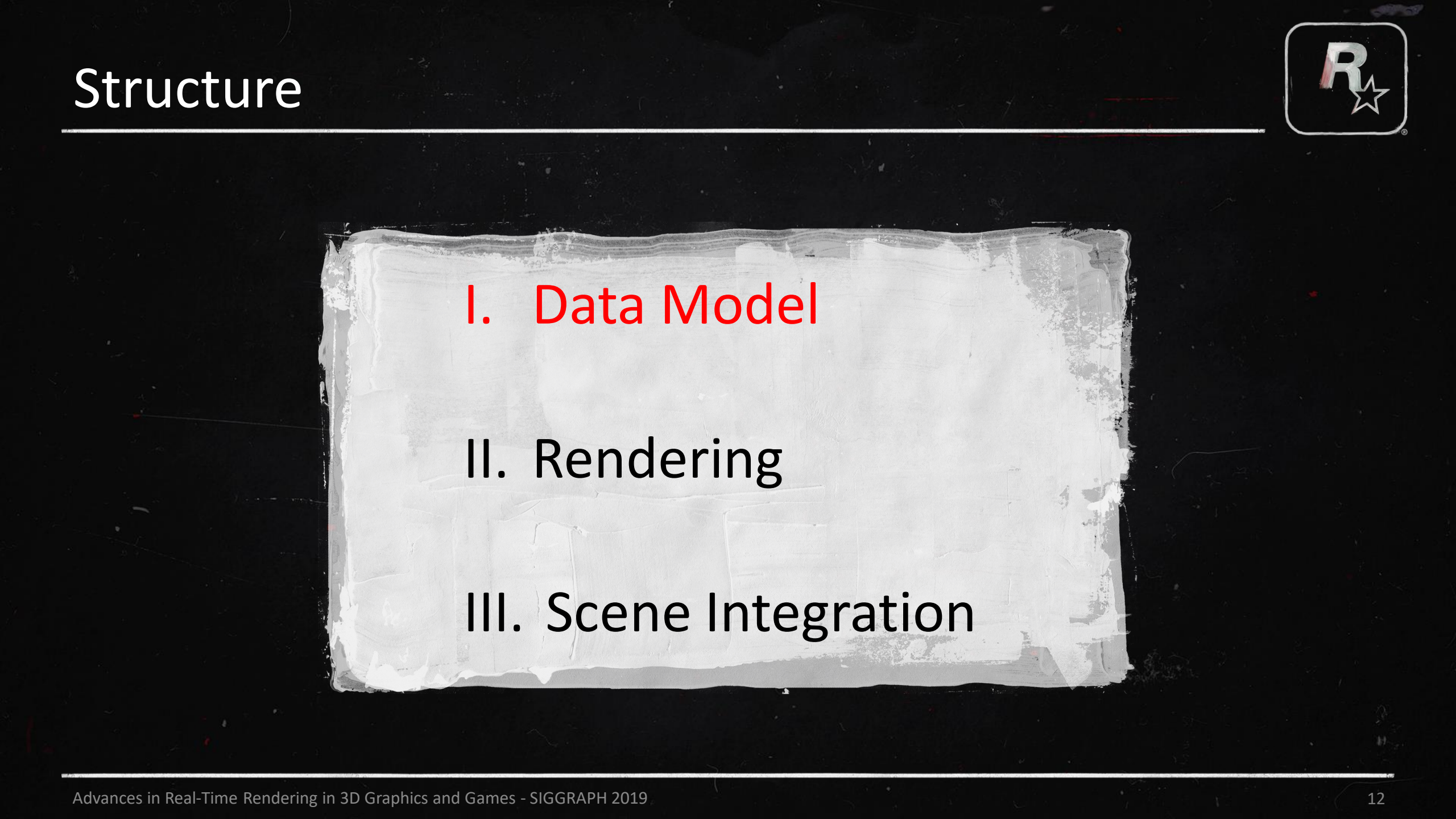


I. Data Model

II. Rendering

III. Scene Integration

Structure

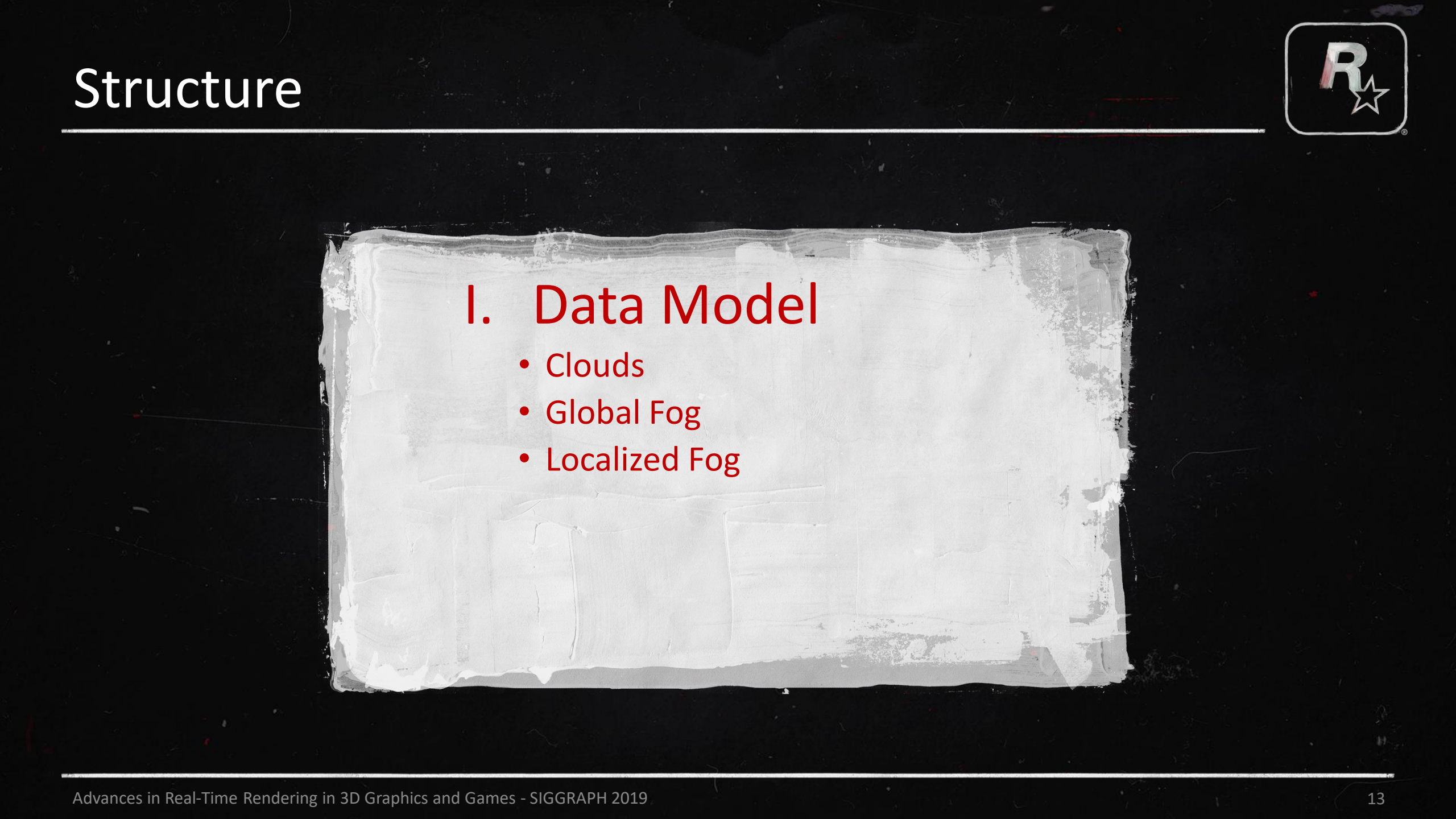


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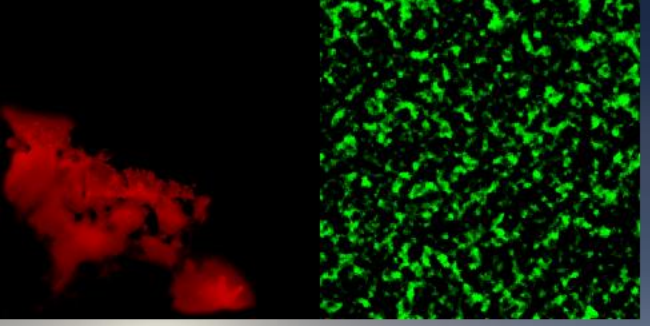
I. Data Model

- Clouds
- Global Fog
- Localized Fog

Cloud Map

- Cloud dome covers world map (~32x32 km^2)
- Modulated by texture and weather data
- 512x512 R16G16 Texture
  RG: Cloud layer 1 + 2

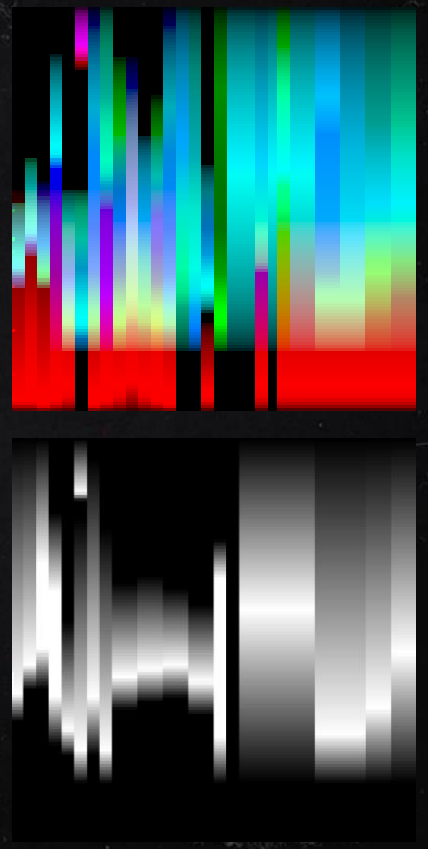




Cloud Map

Cloud Height LUT

- Cloud map for 2D placement
- LUT defines altitude-based shape
  - (Alto)Cumulus and stratus
- Copied from atlas into 1x128 RGBA8 texture
  - Slice determined by weather parameters

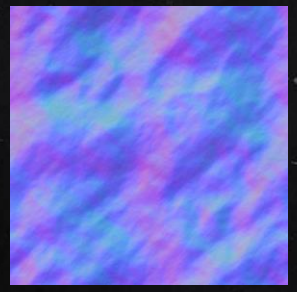


Cloud LUT atlas

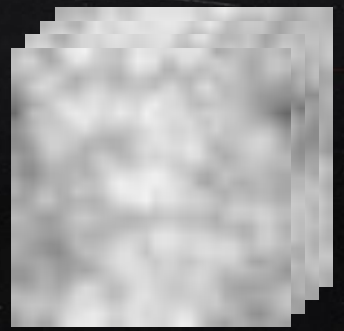
Cloud Detail



- Cloud detail carved out of low frequency shape [Schneider15]
- 2x 2D displacement samples
  - At xy and xz coordinate of sample position
- 1x 3D Perlin/Worley noise samples
  - Displaced via 2D noise + wind offset



2D displacement



3D Perlin/Worley noise




```
float2 c = SampleCloudMap(ray.p);
float3 cloudLut = SampleCloudLUT(altitude);
float density = smoothstep(g_CloudShape.xz + cloudLut.xy, g_CloudShape.yw + cloudLut.xy, c.xy);
```

```
float rescale(vMin, vMax, v)
{
  return saturate((v - vMin) / (vMax - vMin));
}
density = rescale(noise, CloudLut.z, density);
```

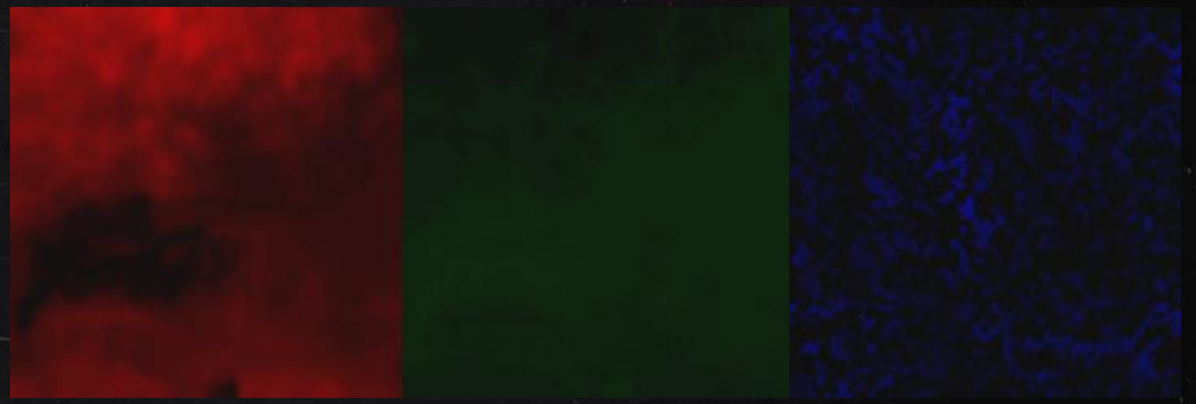
Rain FOG



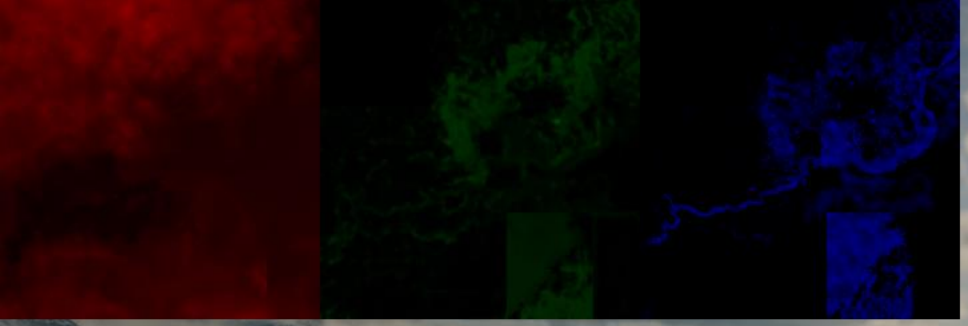
- Rain fog below rain clouds
- Derived from cloud map
- Height based falloff

Fog Map

- Covers playable map (~8x8 km^2)
- 512x512 R11G11B10 Texture
  - R: Fog start height
  - G: Fog falloff distance
  - B: Density



- Generated in composition pass
- Weather and Time Of Day dependent e.g. morning mist



Fog Map

Global height based fog + fog map

Fog Volumes

- Artist-placed
- Attached to particles and vehicles
- Sphere or box shape
- Additive or alpha blending

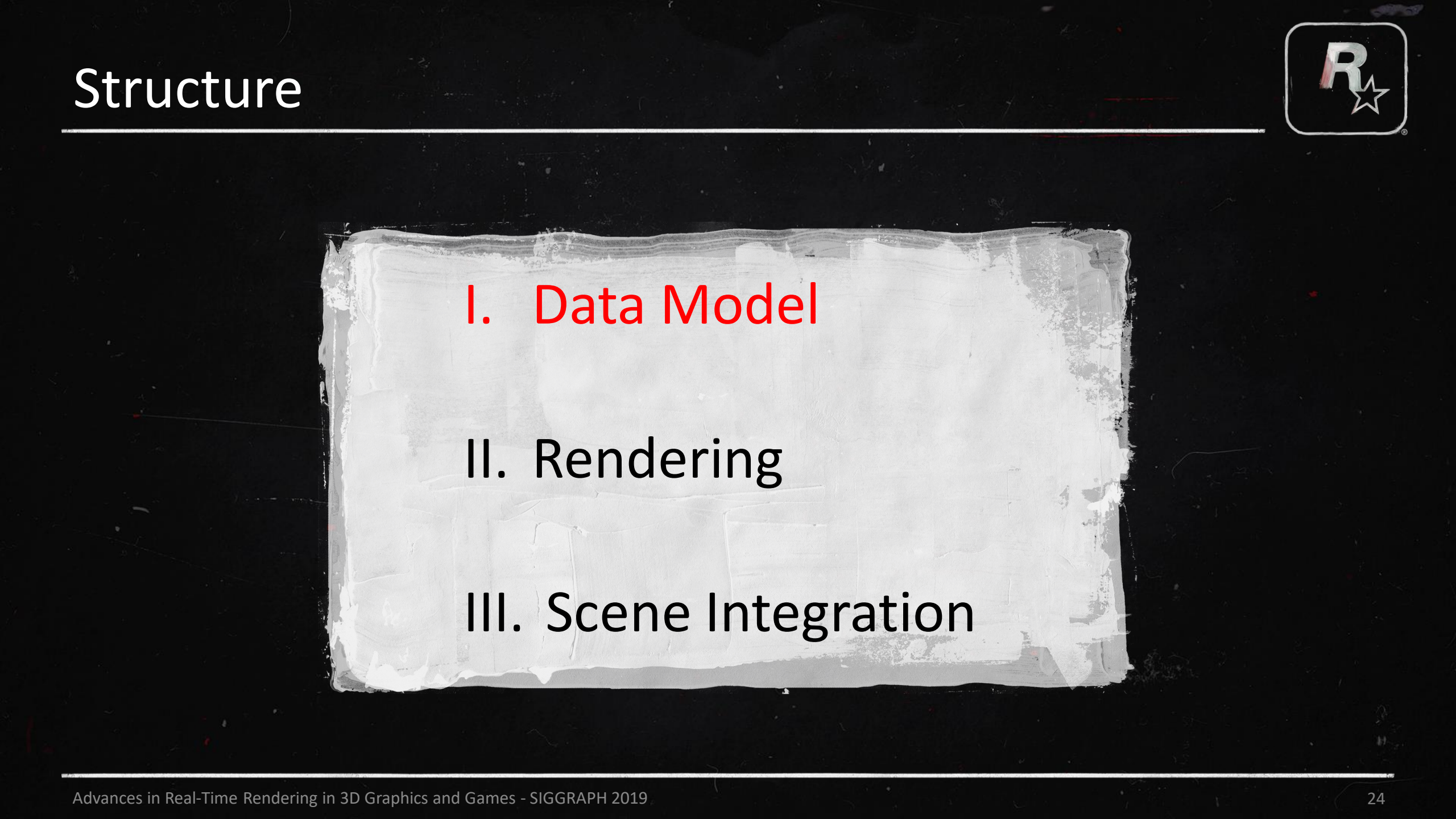




Fog Volumes

Exterior (additive) volume

Structure

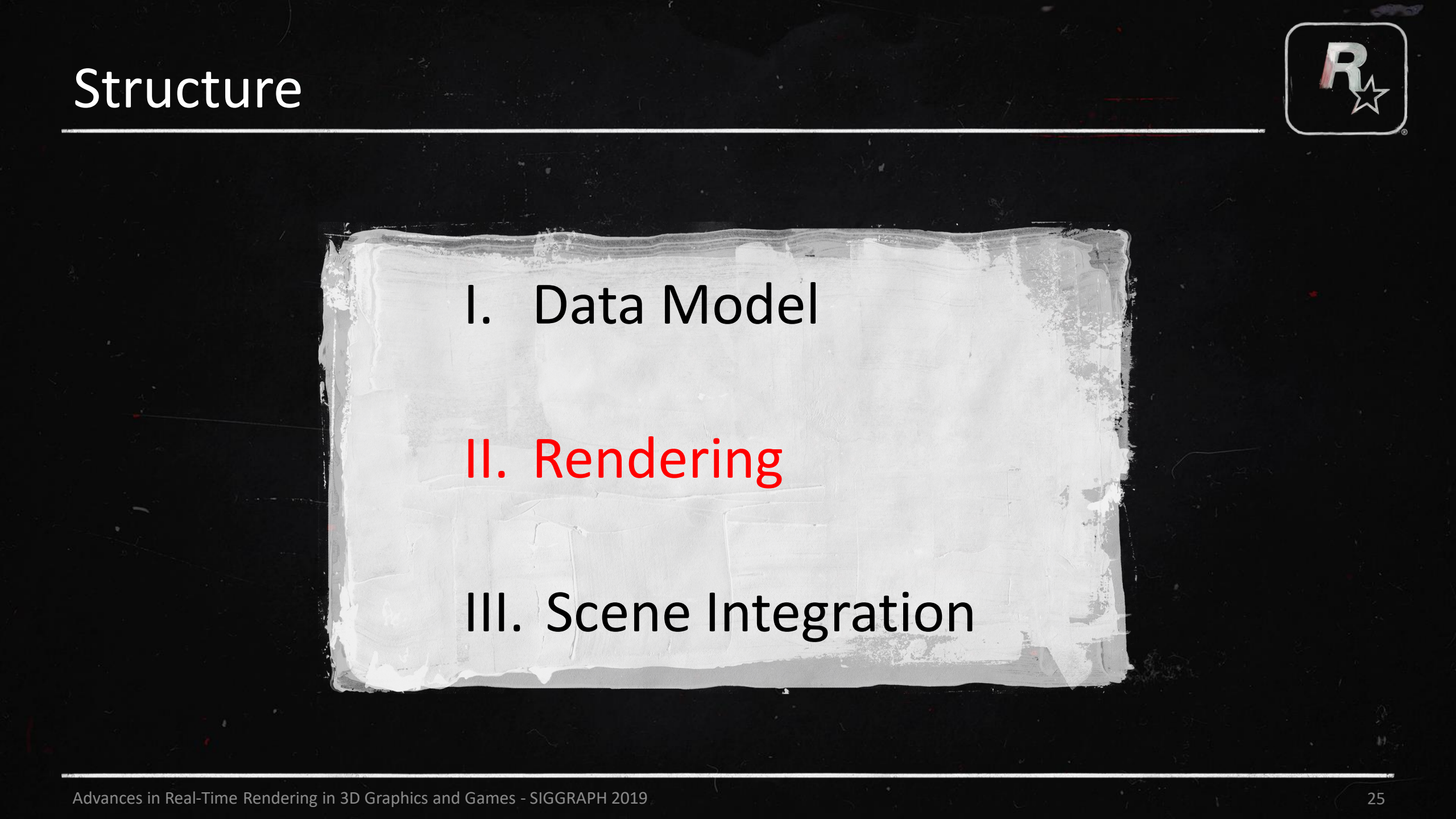


I. Data Model

II. Rendering

III. Scene Integration

Structure



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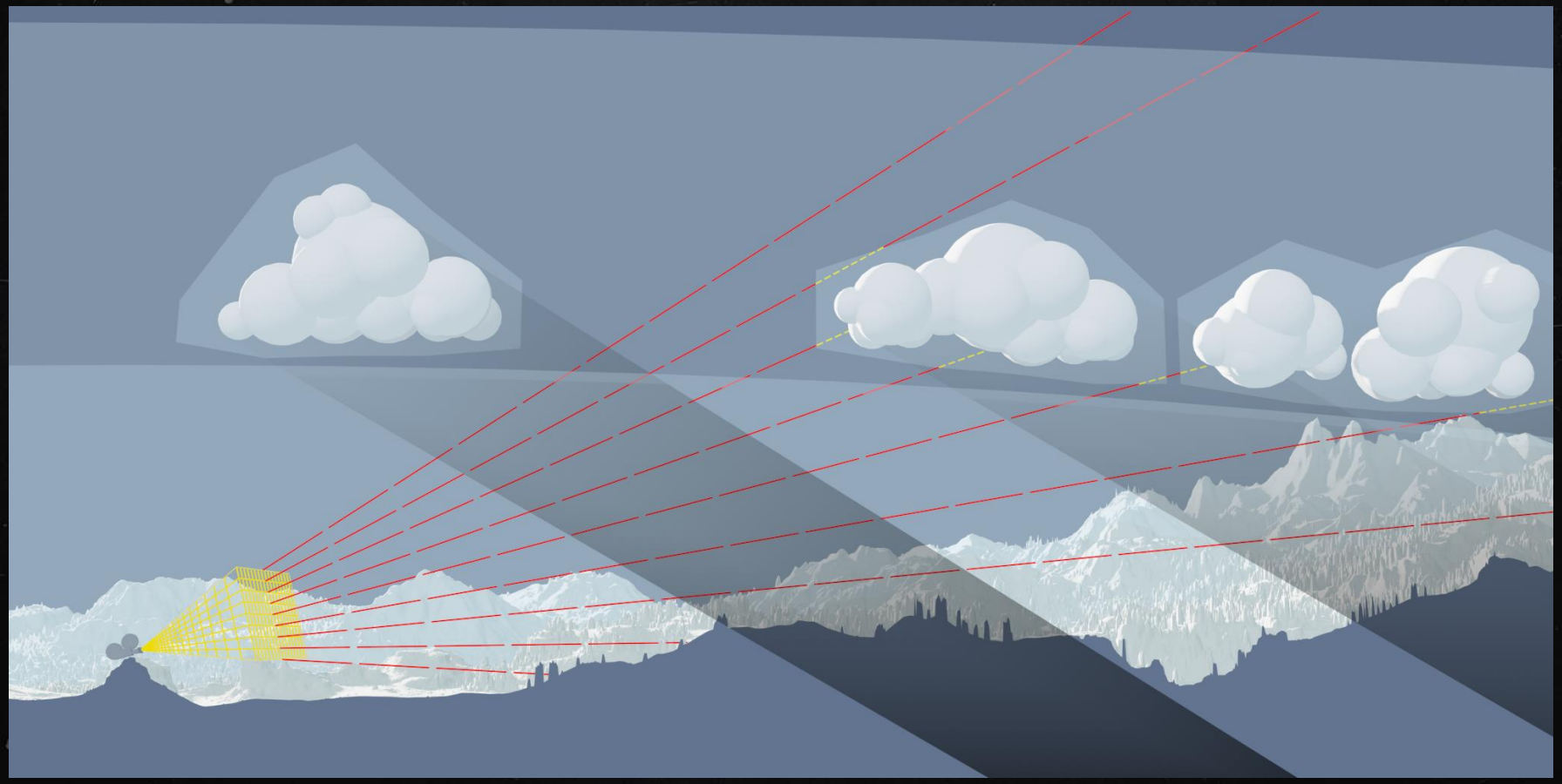
Structure

II. Rendering

- Scattered Light
- Frustum Volumes
- Raymarch
- Performance

Rendering

In a nut shell: Mix of voxelization for near and raymarching for far distance



Shading Model



- Single scattering, transmittance wave length independent
- Two direct light octaves to approximate multi scattering [Wrenninge13]
- Scattered Light SL at x with phase function P and luminance L

$$SL(x) = \sum_{i}^{|lights|} P(g, \theta_i)\, V(i, x) L(i, x)$$

- Visibility V via shadow map lookups and/or secondary ray T

Phase Function



- Artistic liberties + performance considerations
- Modelled around Henyey-Greenstein phase function HG
- Composition of M lobes using artist weights w_0 and w_1

$$HG(g,\theta) = \frac{1}{4\pi}\frac{1-g^2}{(1+g^2-2g\cos\theta)^{3/2}}$$

$$P(g,\theta,\sigma_{ext}) = HG(g,\theta)\ w_0 + \frac{w_1\sigma_{ext}}{M-1}\sum_{j=1}^{M} HG\left(\left(\frac{2}{3}\right)^j g,\theta\right)$$

Settled with $M = 2$

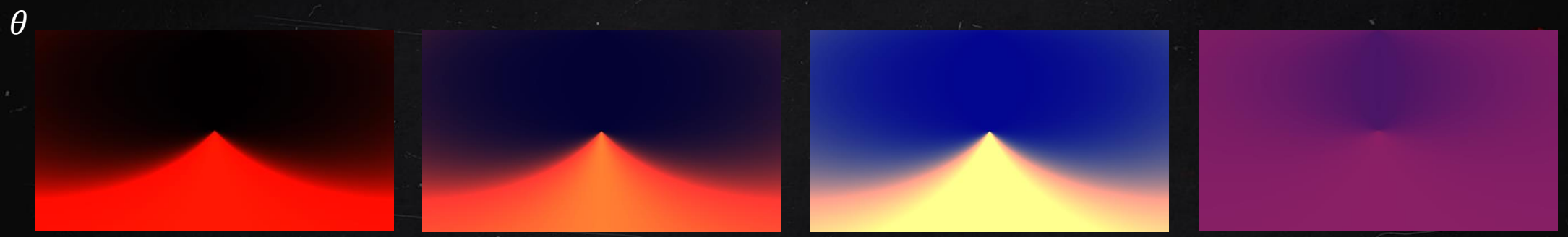
Phase Function



- Still lacks plausible back scattering effect
- Faked by clamping to $\frac{1}{\pi}$ from Lambertian BRDF
- Not physically based
- Based on σ_{ext} and perceptual tuning

```
float ApplyBackScattering(float phase, float extinction)
{
  float v = 1/PI * rescale(BACK_SCATTER_MIN, BACK_SCATTER_MAX, extinction);
  return max(phase, v);
}
```

Phase Function



Low Density Fog
g = 0.9

High Density Fog
g = 0.9

Cloud
g = 0.8

Smoke
g = 0.1

Phase octave 0
Phase octave 1
Back scatter clamp


Phase Function
Phase octave 0 + 1 + back scattering

Visibility

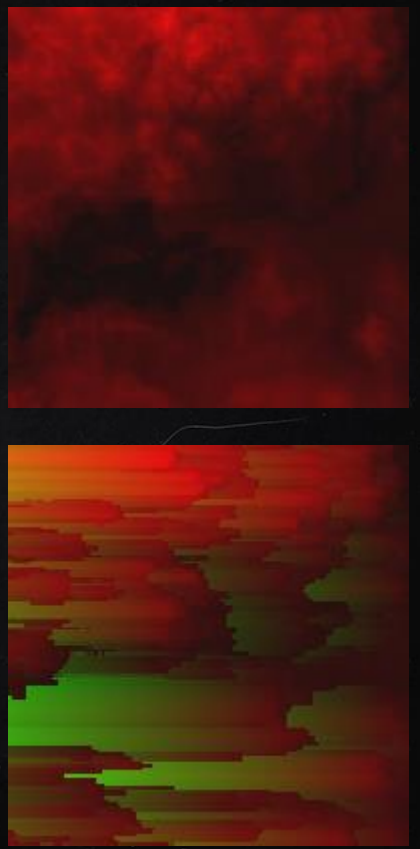


- Shadow map lookups
  - Cascade + local lights
  - Terrain shadows
  - Cloud shadows

- Single transmittance lookup towards sun/moon
  - On cloud samples only
  - Variable distance

Visibility: Terrain Shadow Map

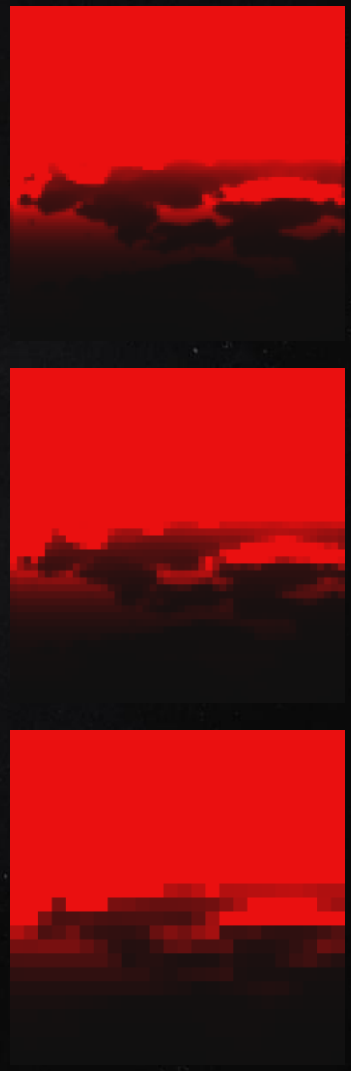
- Raymarch across terrain height map
- 128x128 RG16F
  - R: Intersection height
  - G: Ray length
- Ray length adjusts falloff for soft shadows



Visibility: Cloud Shadow Map

- Stores directional light shadows
  - Exponential Shadow Map [Annen08]
- Raymarch cloud data
  - Store transmittance weighted depth
- 768x768 R16F, 6 mips
- Blur + mip chain to reduce aliasing/crawling

```
float transmittance = 1;
float depth = maxDepth;
while (ray.t < maxDepth && transmittance > 0.001)
{
  float density = SampleCloudDensity(ray.p + ray.d * ray.t);
  transmittance *= exp(-density * stepLength);
  depth = lerp(depth, ray.t, pow(transmittance, 4));
  ray.t += stepLength;
}
```



Ambient Light

- Visibility: Reuse directional light secondary ray sample on clouds

- For raymarching: Average sky ambient without clouds
  - Sky scattering rendered into low resolution paraboloid texture
  - Parallel reduction
  - Samples weighted towards sun/moon direction

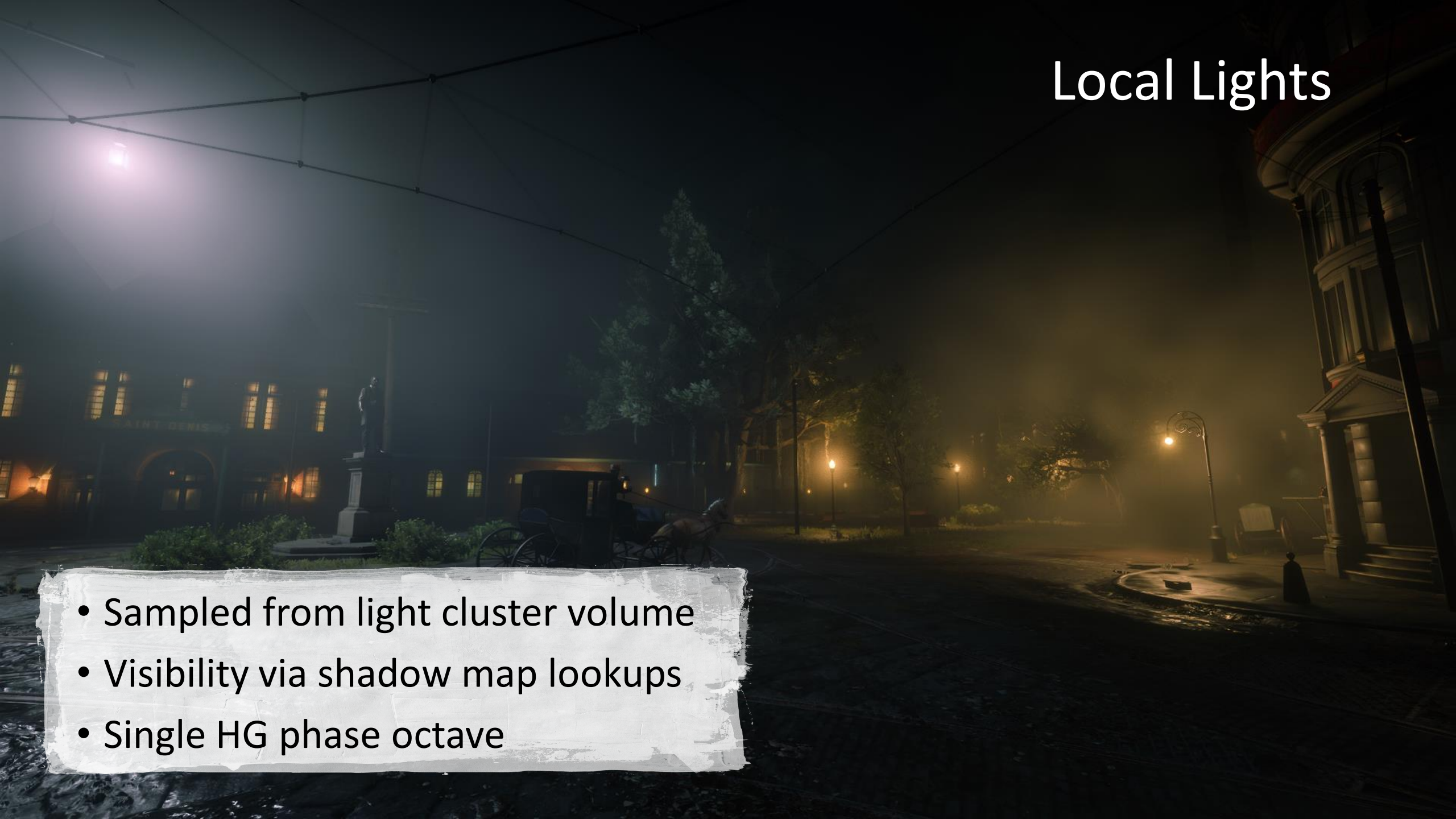


- For frustum volume: Irradiance probe field
  - Sky light + indirect bounce light * AO



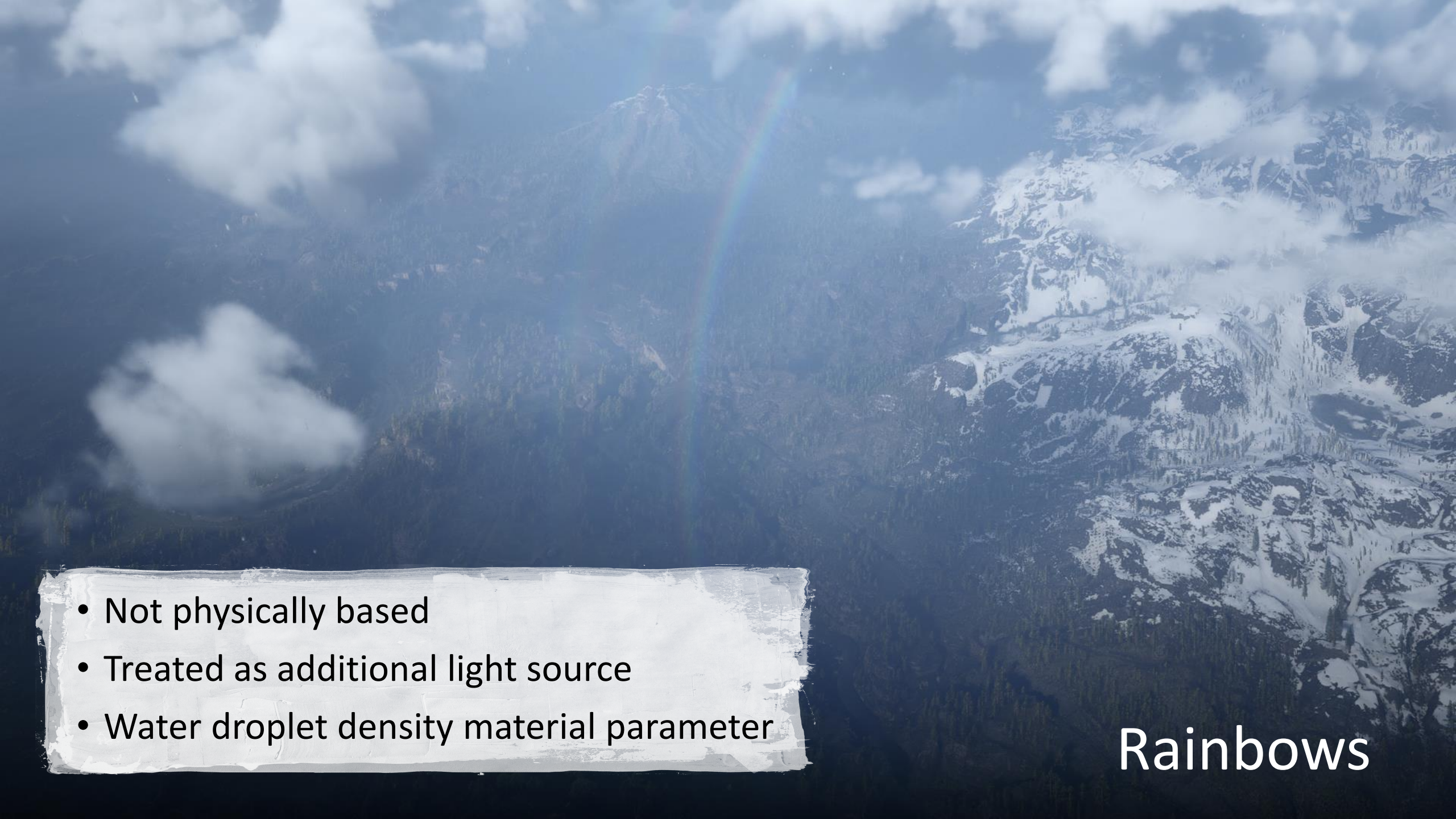
Ambient Light

Irradiance probes + AO



Local Lights

- Sampled from light cluster volume
- Visibility via shadow map lookups
- Single HG phase octave



Rainbows

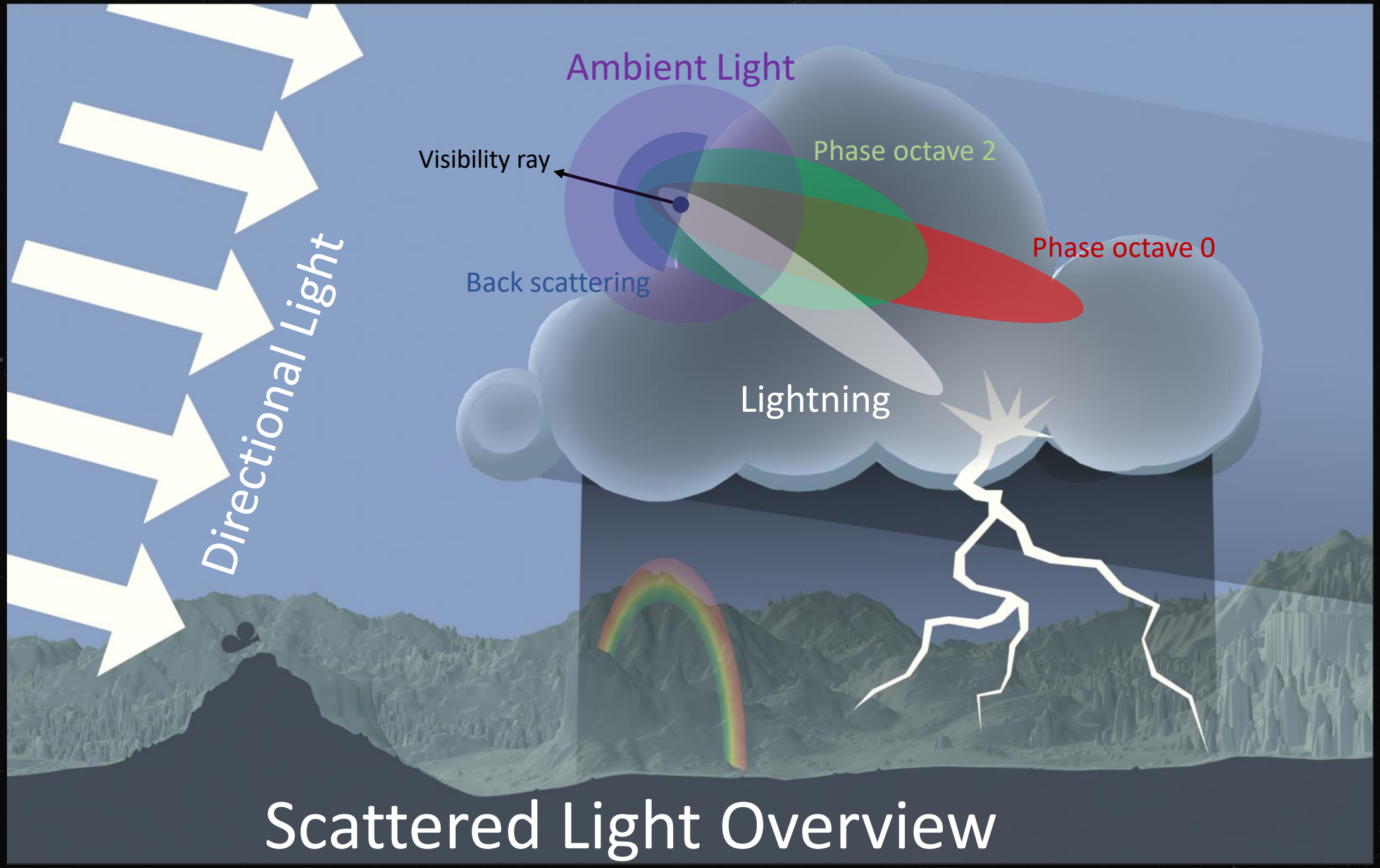
- Not physically based
- Treated as additional light source
- Water droplet density material parameter

Lightning

- Lightning strikes populate point light array

- Loop over array for each sample step
  - Exponential falloff + HG phase function
  - Simplified local extinction
  - Added to scattering term

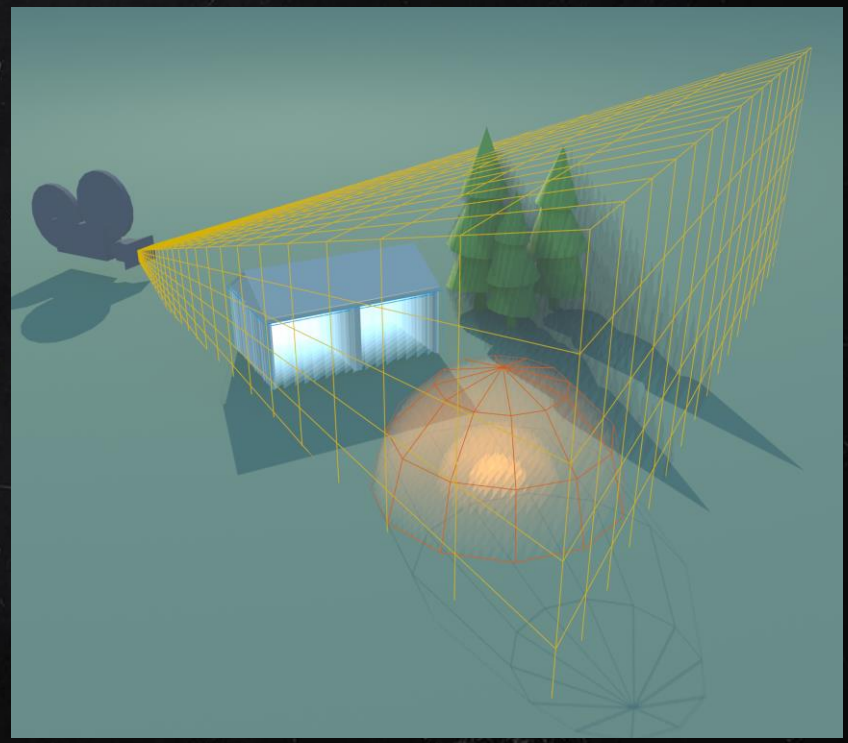




Scattered Light Overview

Frustum Voxel Grid

- Based on Volumetric Fog [Wronski14] and PBR-based Volumetric Lighting [Hillaire15]
- Dynamic depth coverage (<160m)
- Divided into 3 clip-space volumes
  - Shadow Volume
  - Material Volume
  - Scattered Light and Extinction Volume
- Resolution of 160x88x64



Shadow Volume

- Format: R16F
- Contains directional light shadow term
  - Shadow map cascade
  - Cloud shadow map
  - Terrain shadow map
- Temporal filtering to improve stability



Material Volume

- 2x RGBA8UB for material parameters
  - Volume A
    - RGB: Scattering σ_{scat}
    - A: Absorption σ_{abs}
  - Volume B
    - R: Phase g
    - G: Global emissive intensity
    - B: Ambient intensity
    - A: Water droplet density



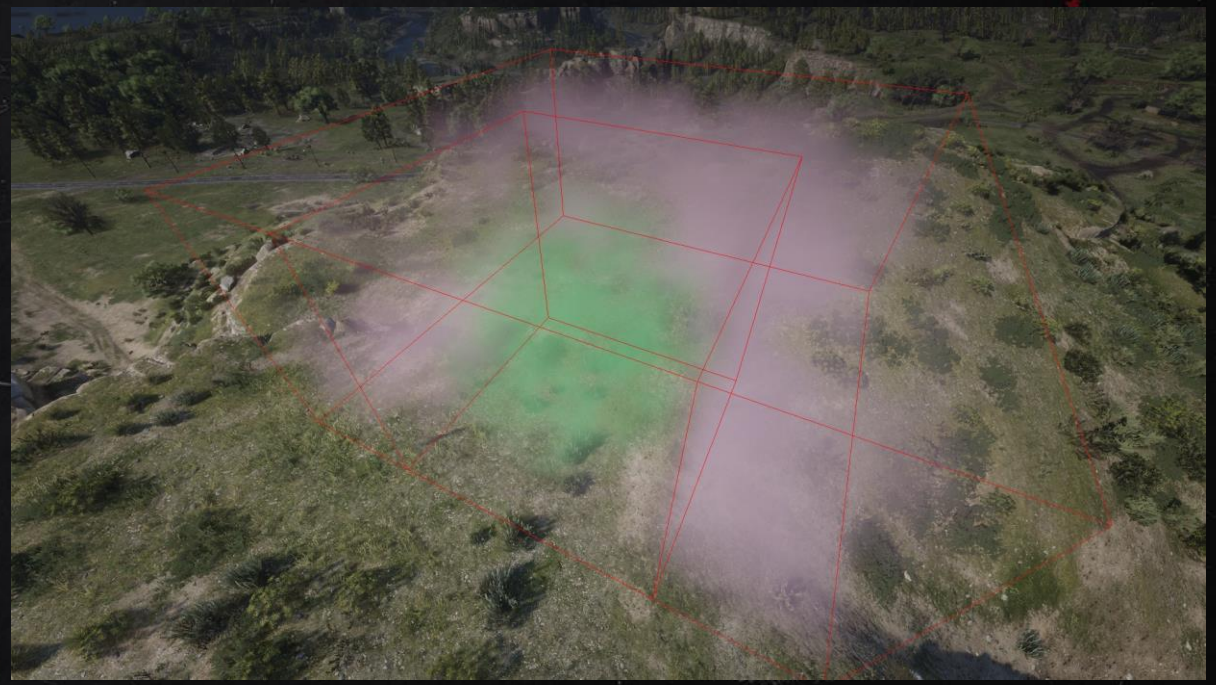
- Up to 32 volumes stored in cluster grid
  - Z-Binning [Drobot17]

Material Volume

- Samples and blends all material sources

Blend order:

Additive => Alpha => Particles



- Individual tiling noise samples

Material Volume



- Temporally filtered (volume A only)
- Wind velocity as motion vector
- Reprojection velocity magnitude volume, R8UB
  - Modulates motion vector
  - Used to avoid leaking issues in interiors

Scattered Light Volume

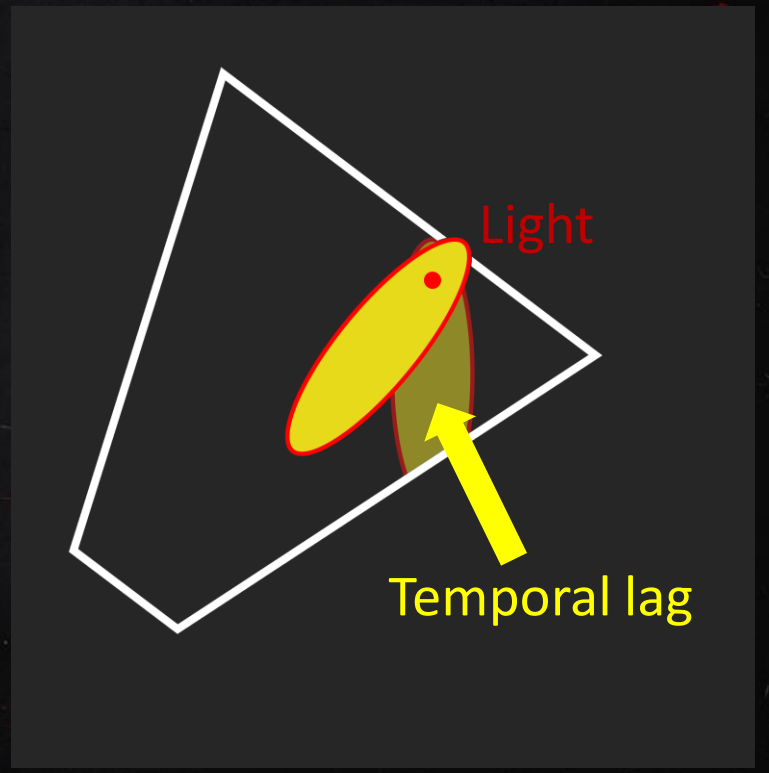


- Samples material and shadow volume
- Direct light + ambient light (irradiance probes)
- RGBA16F
  - RGB: Scattered light
  - A: Extinction
- Slice march for in-scattering + transmittance integration in accumulation pass

Scattered Light Volume

NO temporal blending on scattered light volume

- Light ghosting with moving lights
- Trailing with strong forward scattering on moving camera
- Dithered lookups + TAA to stabilize instead



Frustum Volumes

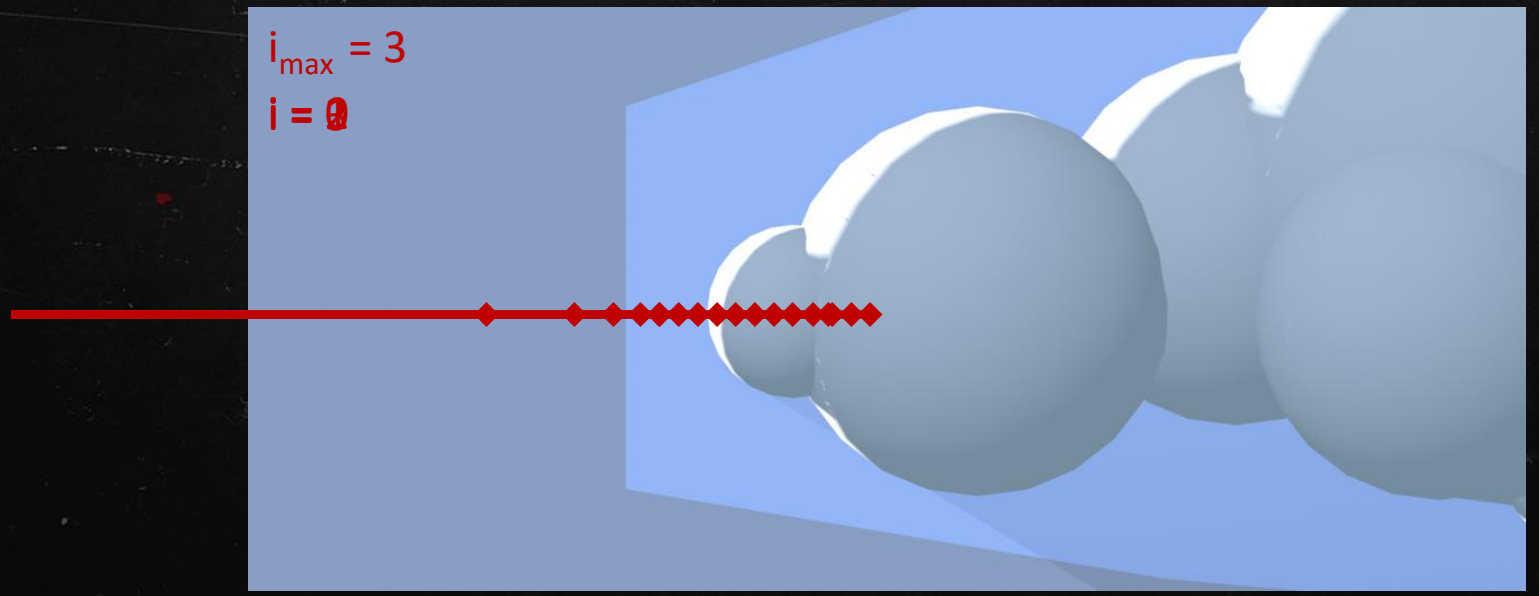


- Undersample data at large distances
  - Limited resolution
  - Exponential slice distribution
  - Unstable, even with temporal integration

- Need a system to “take over” at some distance
  - Raymarching
  - Can ignore fog volumes, irradiance probes and local lights

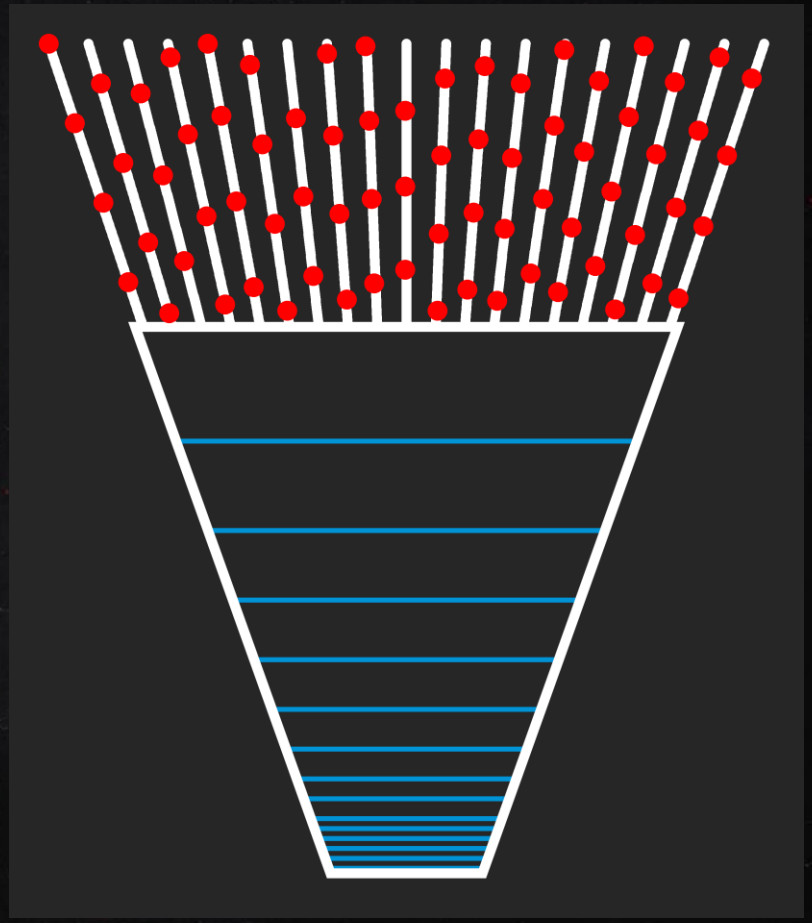
Raymarch

- Ray length defined by screen depth, ground plane and cloud dome
- Iterative refinement of ray step t_{base}: $t = t_{base} 0.5^{i}$
- More samples on cloud material [Schneider15]
- Can potentially jump over thin cloud
  - Tune base step length and data



Raymarch: Main Viewport

- Tracing starts where voxels end
- Ray origin offset using blue noise
- Half res target resolution
  - Sufficient for our data
  - Still expensive
  - Reconstruction over 4 frames

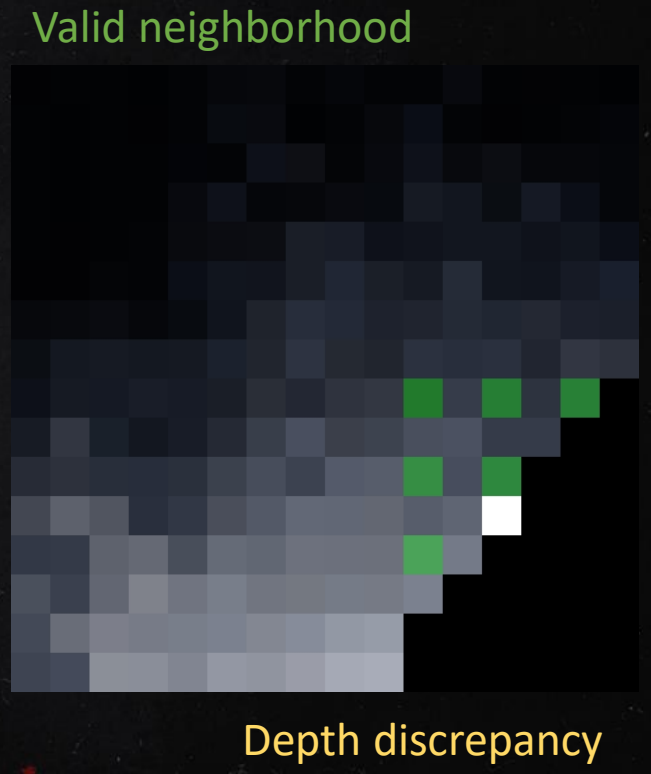


Raymarch Reconstruction

- Cast 1 out of 4 rays

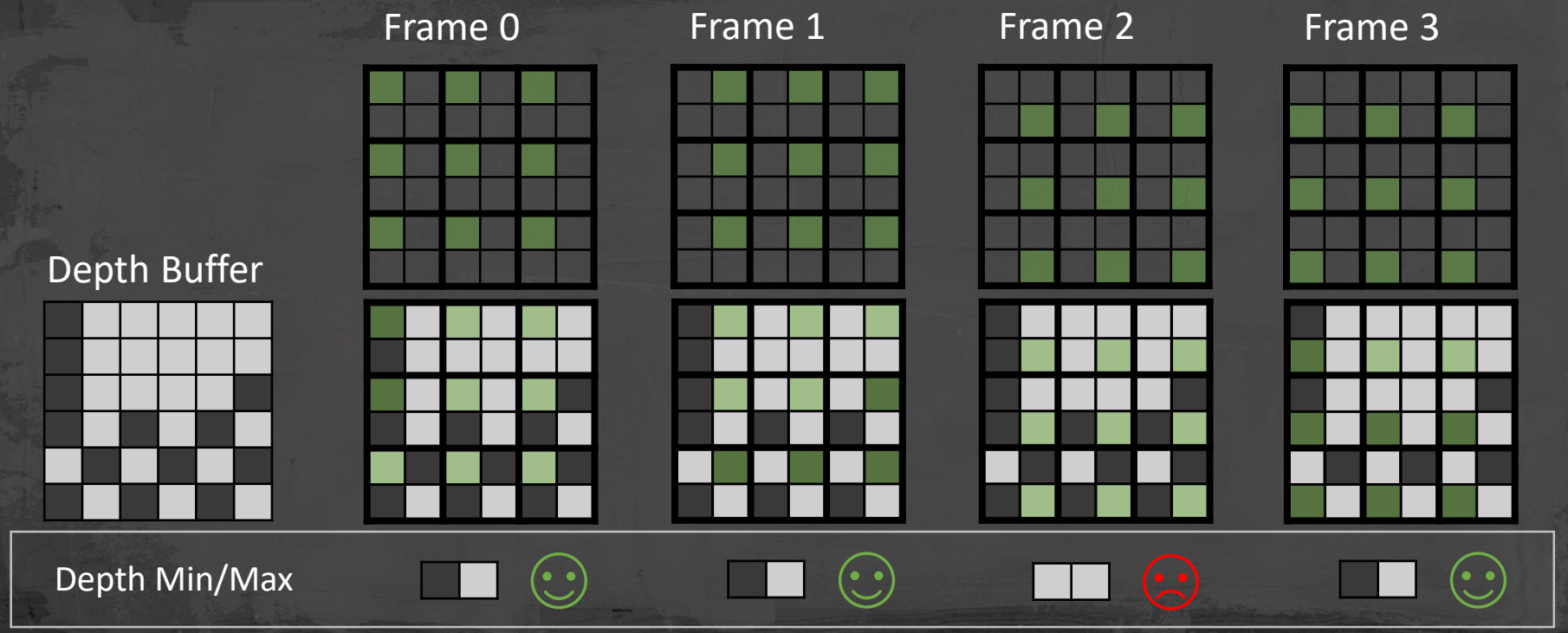
- Reproject to previous frame for skipped pixels
  - Color AABB clamp from 3x3 neighborhood
  - Exclude neighborhood pixels with high depth distance to current pixel
  - No history? Use nearest depth sample from neighborhood

- Cohesive ray placement at depth discrepancies
  - Checkerboard min/max depth downsampling
  - High depth variance hard to resolve (i.e. vegetation, hair, stochastic transparency, etc.)



Raymarch Reconstruction: Ray Placement

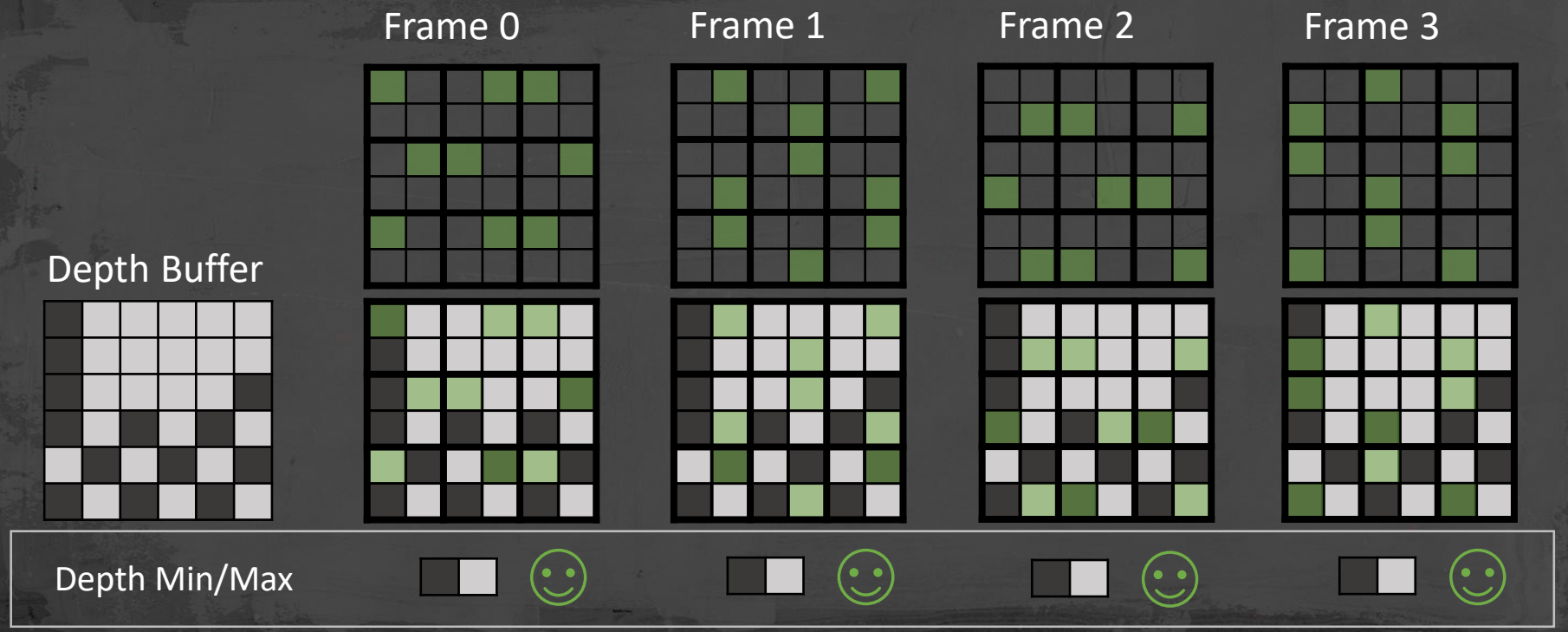
- Goal: Cohesive depth coverage
- First attempt: Uniform ray position, based on current frame index (0-3)



Long rays only:
No up-to-date information for near depth in this tile

Raymarch Reconstruction: Ray Placement

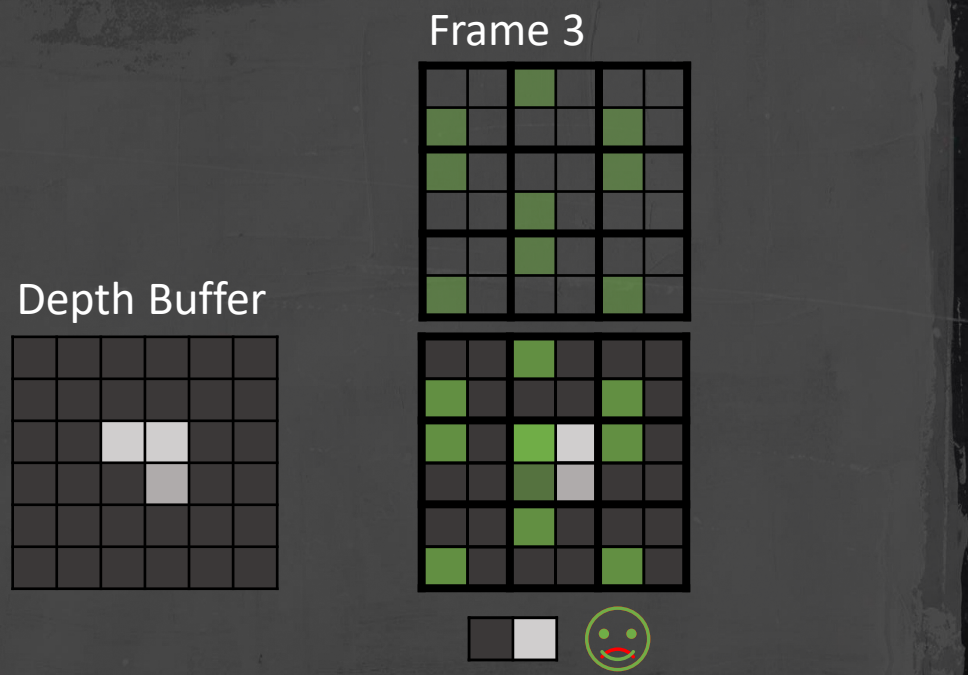
- Second attempt: Start with frame index, add +1 offset in checkerboard pattern





Raymarch Reconstruction: Ray Placement

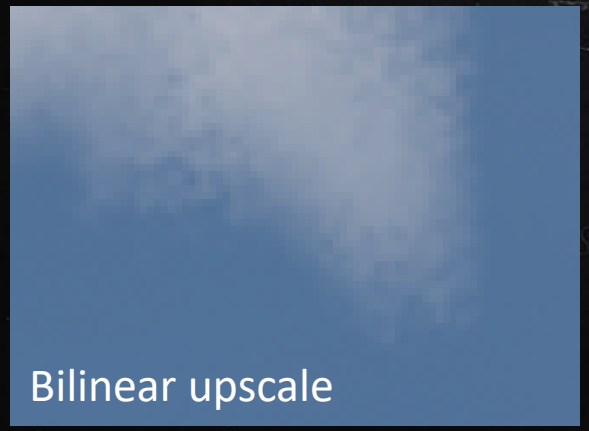
- Can still go wrong in some cases
- Third Attempt: Checkerboard offset + depth neighborhood analysis
- Compare depth min + max of current 2x2 tile with eight neighbor tiles



Place ray at location with under represented depth

Raymarch: Full Resolution Upscale

- Still need to upscale half res to full res
- Dithered upscale with blue noise offsets [Stachowiak15] [Gjoel16]
  - Four taps, chose closest depth
  - Kernel width based on weighted ray depth







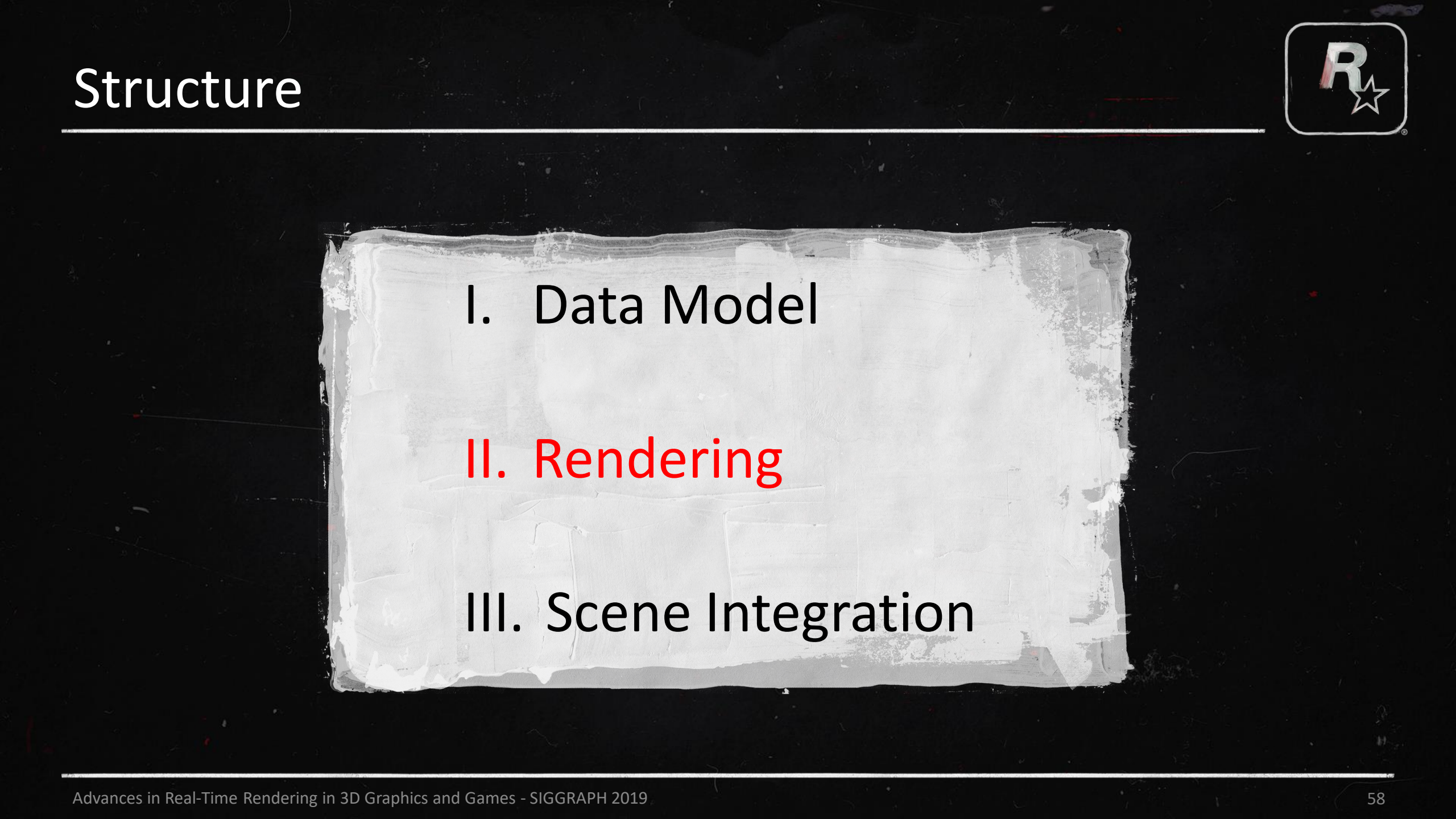
Performance

- Heavily dependent on cloud cover and fog composition
- Artists responsible too!
- Worst offender: Long rays
- Implementation fully compute-based
  - Leverage async compute
  - Overlap dispatches whenever possible
- Time slice terrain and cloud shadows

Cloud/Fog Map	Cloud ESM	Shadow Volume	Material Volume	Scattering Volume	Raymarch	Raymarch Reconstruction
0.04ms	0.156ms	0.204ms	0.33ms	0.34ms	1.31ms	0.232ms

Measured on PS4

Structure

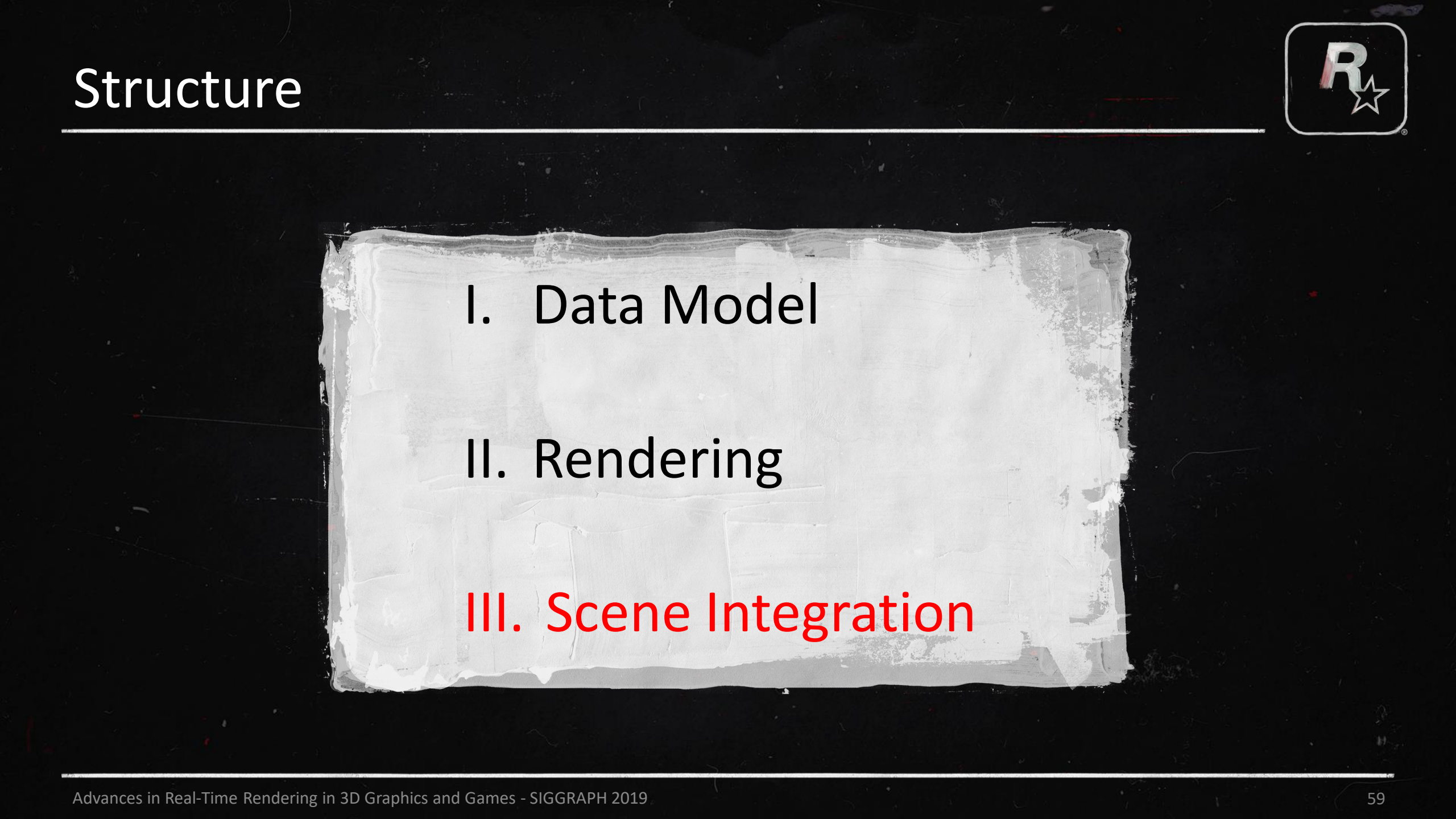


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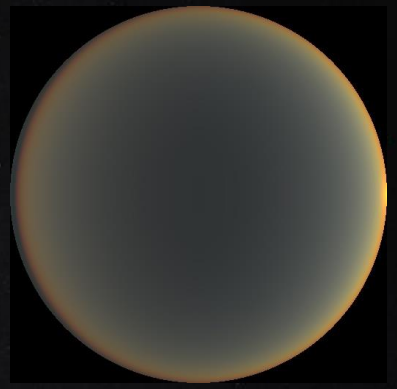
Structure

III. Scene Integration

- Main Viewport
- Irradiance Probes
- Directional Light
- Reflections

Main Viewport: Sky Scattering

- Precomputed Atmospheric Scattering [Bruneton07]
  - Improvements from [Hillaire16]
  - Supports earth shadow
- Time-sliced update
- 32x16x32 frustum volume
  - LUT for forward and deferred lighting



+ earth shadow



Main Viewport: Sky Scattering



- Sky scattering at each sample point?
  - Very expensive
  - Instead applied once at weighted depth [Hillaire16]
- Visibility
  - Separated from integration
  - Accumulate shadowed area
  - Attenuated by transmittance



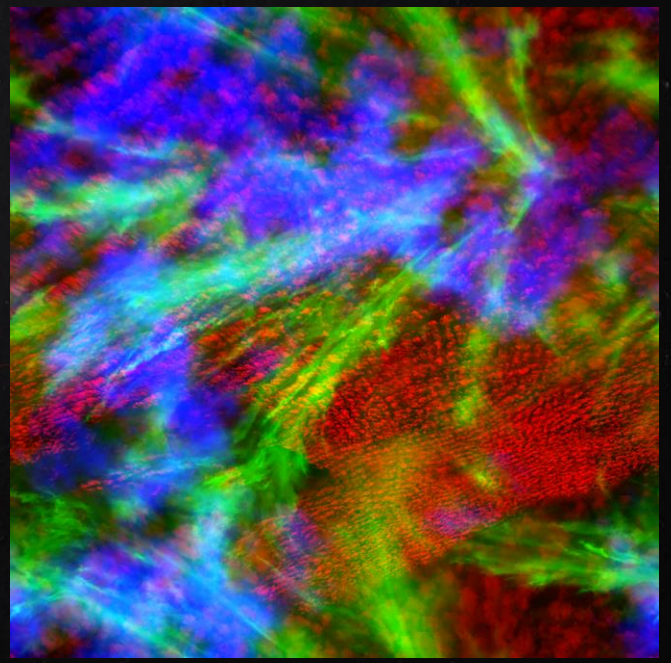
Main Viewport: Sky Scattering

Final Composition

Scene Integration: Main Viewport



- Deferred application
  - Volume lookup
  - Merged with raymarch result lookup
  - Merged with sky scattering and transmittance
- Cirrus cloud layer
  - ~8 km altitude
  - Single 2D texture lookup for density
  - Same shading model as before



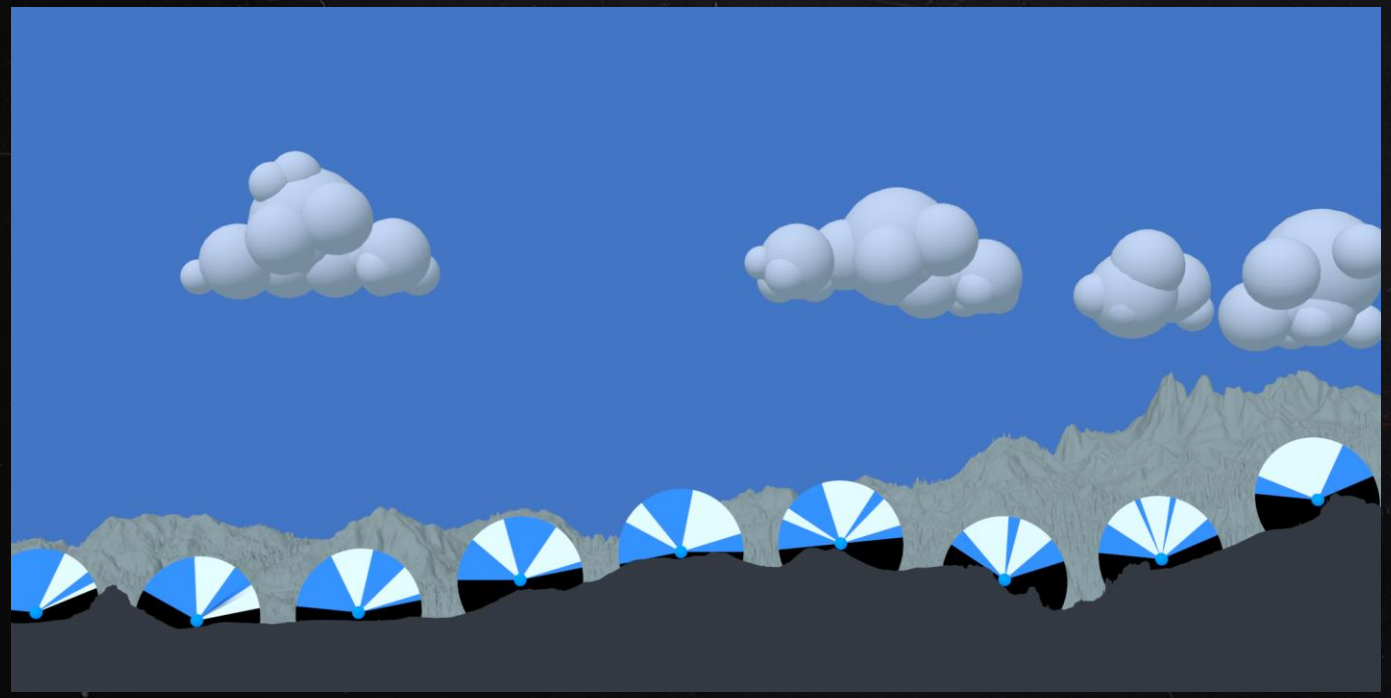


Scene Integration: Main Viewport

Cirrus + Raymarch + Volume + Sky Scattering/Transmittance

Scene Integration: Sky Irradiance Probes

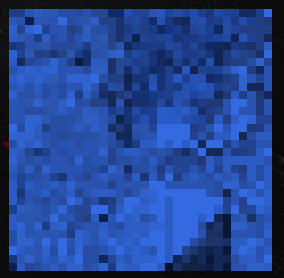
- Sky irradiance probe map (32x32): 256x256m^2 per probe
- Encoded as 3rd Order Spherical Harmonics
- Position/occlusion prebaked



Sky scattering paraboloid



SH Band 0



Clear



Overcast

Scene Integration: Sky Irradiance Probes



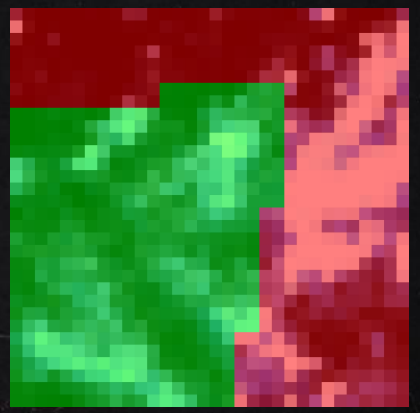


Sky irradiance with scattering/transmittance and direct light

Scene Integration: Sky Irradiance Probes



- Basis for higher resolution probe grid
- Optimization
  - Large ray steps, reduced detail
  - Classification pre-pass
  - Cast 32 rays if visible, 16 otherwise
  - One ray/direction per thread
  - Shared memory parallel reduction
- ~0.22ms on PS4





Scene Integration: Directional Light

Cloud shadow map sampled

Scene Integration: Reflection Probes

- Cube map at camera position responsible for majority of reflections
- Low res cube map for scattering/transmittance
  - Simplified raymarch
  - Needs to match sky irradiance probes!
- ~0.15ms on PS4



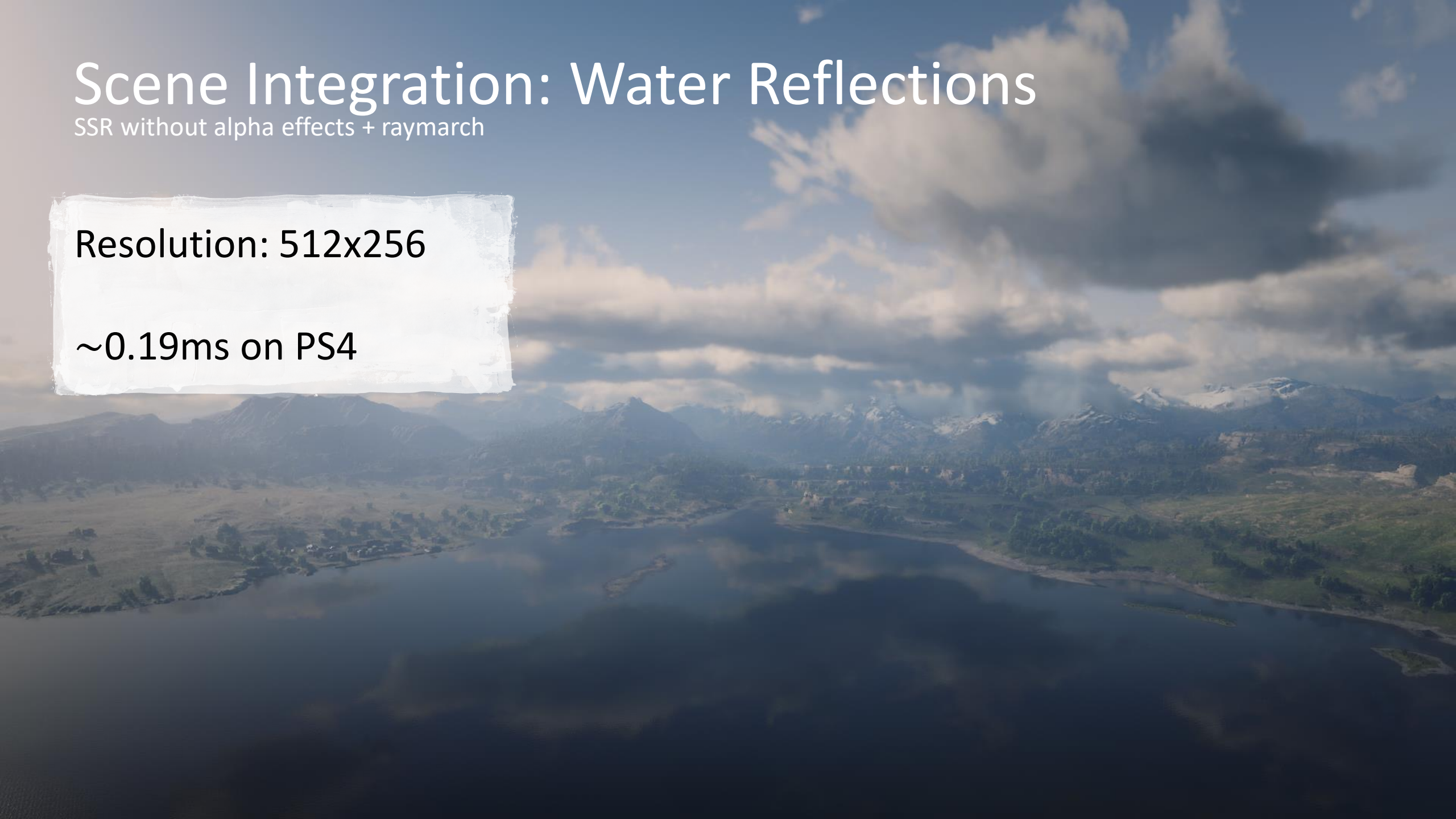
Scene Integration: Water Reflections



- Mix of SSR, reflection probe, planar reflections and height map tracing
- SSR without alpha/volumetrics
- Dedicated raymarch pass
  - Simplified + heavy temporal blend

Scene Integration: Water Reflections

SSR without alpha effects + raymarch



Resolution: 512x256

~0.19ms on PS4

Summary



- Volumetric effects as first class citizens
- Unified, physically-based, multi-material scattering and transmittance system
- Frustum-aligned volume for near distance
  - Fog volumes, local lights and irradiance probes
- Raymarch for far distance
  - Temporal reconstruction
  - Reused and customized for indirect diffuse and specular

Future Outlook



- More material variety
- Wavelength dependent transmittance
- Volumetric shadows
- Localized weather
- Improved denoising

Acknowledgements



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- The Rockstar graphics and lighting team for their collaboration

- Natalya Tatarchuk

…and all of Rockstar Games

https://www.rockstargames.com/reddeadredemption2/thankyou

Thank You



WE’RE HIRING

https://www.rockstargames.com/careers/

Questions?



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